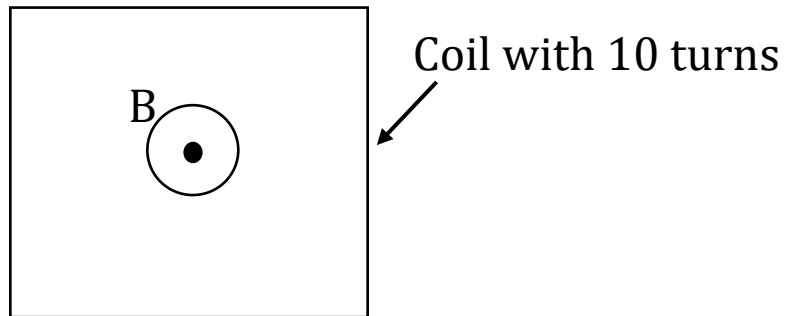


ECE 3113 - Energy Conversion

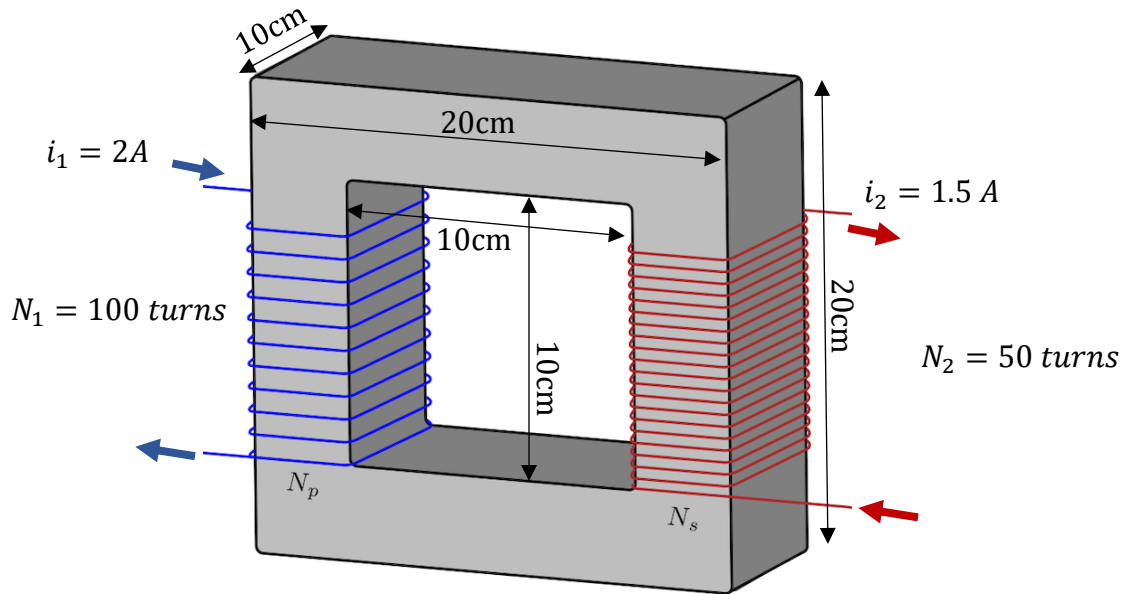
Homework 2

(Due on Monday 10/02/2023)

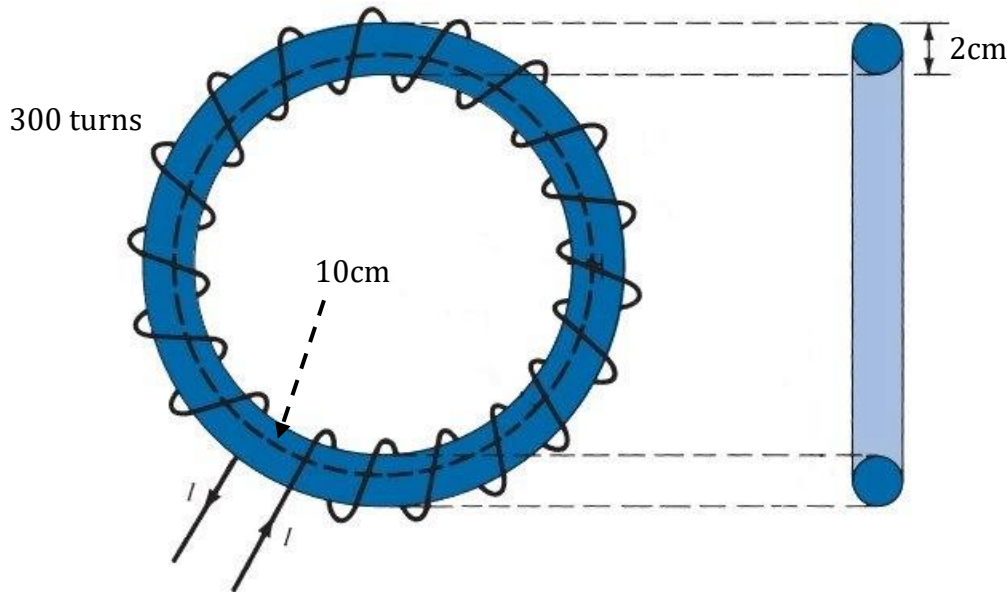
1. Consider a flat square form coil with $N = 10$. The coil dimension is 15 cm on each side, and has a sinusoidal magnetic field of $B = 2 \sin(2\pi 50 t)$ passing through it. The plane of the coil is perpendicular to the magnetic field which points out of the page.
 - (a) Find the induced voltage in the coil. (10 points)
 - (b) If the coil has a magnetic field of 2T and nothing gets changed, what is the induced emf? (10 points)



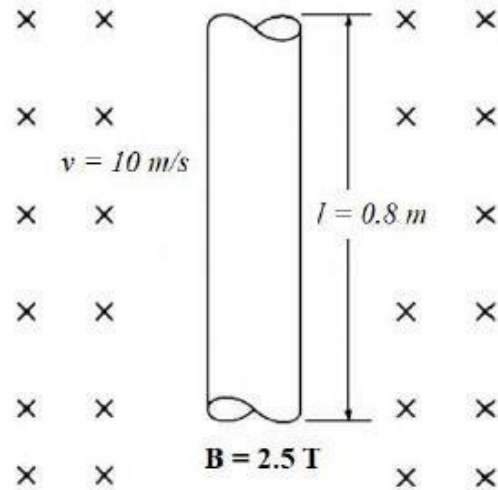
2. Consider the two-legged core shown below. The winding on the left leg of the core (N_1) has 100 turns, and the winding on the right (N_2) has 50 turns. The coils are wound in the directions shown in the figure. If each leg is 20 cm long (measured from outer boundaries) and 10 cm wide, then calculate the flux produced by currents $i_1 = 2A$ and $i_2 = 1.5A$? (Assume $\mu_r = 1000$ and constant) (Core depth = 10 cm). (20 points)



3. An iron rod, 2 cm in diameter and 10 cm long is bent into a closed ring and is wound with 300 turns of wire. It is found that when current of 0.5 A is passed through this coil, the flux density in the coil is 0.5 Wb/m^2 . Assuming that all the flux is linked with every turn of the coil,
- Calculate B/H ratio for the iron. (10 points)
 - What voltage would be developed across the coil if the current passing through the coil is interrupted and the flux in the iron falls to 10% of its former value in 0.001 second? (10 points)



4. Consider that the wire of length $l = 0.8 \text{ m}$, shown below,
 a) is moving in the presence of a magnetic field. Find the induced voltage in the wire; when $v = 10 \text{ m/s}$, $B = 2.5 \text{ T}$ and the moving direction is perpendicular to the wire, as shown. (10 points)



- b) is carrying the 2 A current in the present of the magnetic field with magnetic density of $B = 1.5 \text{ T}$. Find the force induced on the conductor. (10 points)

